

Open-YOLO 3D: Towards Fast and Accurate Open-Vocabulary 3D Instance Segmentation

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DATASET: SCANNET200 (VAL)
mAP 24.7%

SPEED
~16× faster

TIME/SCENE
~22 s

CODE
[GitHub](#)

Preprint • Under review

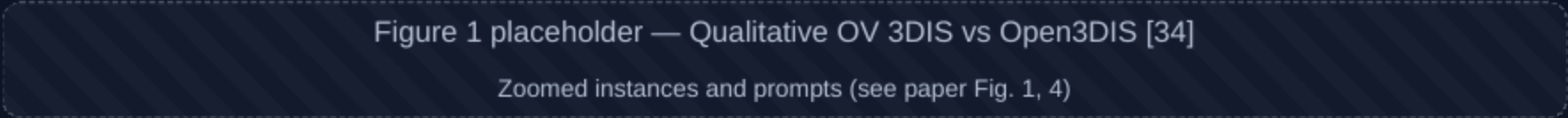
Open-YOLO 3D

ABSTRACT

Fast, accurate Open-Vocabulary (OV) 3D instance segmentation without heavy 2D foundation models

- Replaces Segment Anything Model (SAM) [23]/CLIP [55]-heavy pipelines with Open-Vocabulary Object Detection (OVOD) [7] + class-agnostic 3D masks.
- Generates 3D masks; associates them to text via Multi-View Prompt Distribution (MVPDist).
- Low-Granularity (LG) label maps from 2D boxes; no 2D mask refinement needed.
- State-of-the-art on ScanNet200 [38], Replica [40] with large speedups (~16×).

OV: Open-Vocabulary	SAM: Segment Anything Model	OVOD: Open-Vocabulary Object Detection
MVPDist: Multi-View Prompt Distribution	NMS: Non-Maximum Suppression	DBSCAN: Density-Based Spatial Clustering of Applications with Noise



Why Open-YOLO 3D?

- 3D instance segmentation (3DIS) crucial for robotics/AR; existing OV 3DIS methods are slow (5–10 min/scene) due to SAM [23]/CLIP [55] multi-view lifting.
- Projection of class-agnostic 3D masks already encodes instance regions; SAM adds redundancy.
- Use fast OVOD detectors (e.g., YOLO-World [7]) to provide text-category signals.
- Context: OpenScene [35] and ConceptGraphs [13] lift 2D CLIP to 3D; OpenMask3D [42], OVIR-3D [32], Open3DIS [34] use SAM/CLIP but are slow.

Baselines cited with refs: OpenScene [35], ConceptGraphs [13], OpenMask3D [42], OVIR-3D [32], Open3DIS [34]

Key Contributions

- 2D detection-based OV labeling of 3D instances → efficiency over 2D segmentation.
- Bounding box–only scoring and MVPDist to match prompts to 3D masks.
- SOTA accuracy with substantial speedups on ScanNet200 [38] and Replica [40].

Open-YOLO 3D Overview

- 3D proposal network (Mask3D [39]) → class-agnostic 3D masks.
- 2D OVOD (YOLO-World [7]) → per-frame boxes + class labels.
- Build Low-Granularity (LG) label maps from boxes (size-weighted overwrite).
- Project points to frames, compute per-mask visibility with VAcc, pick top-k views.
- MVPDist aggregates per-point labels across views → final class.

Figure 2 placeholder — Pipeline diagram

3D proposals + 2D OVOD → LG maps → VAcc → MVPDist

3D Proposals and LG Label Maps

3D Object Proposals (Mask3D [39])

- Hybrid CNN + transformer mask decoder.
- Fast class-agnostic masks vs 2D-fusion proposals [32, 34].

Low-Granularity (LG) Label Maps

- Initialize -1; overwrite with box label by descending area.
- Efficient; no SAM masks required [23].

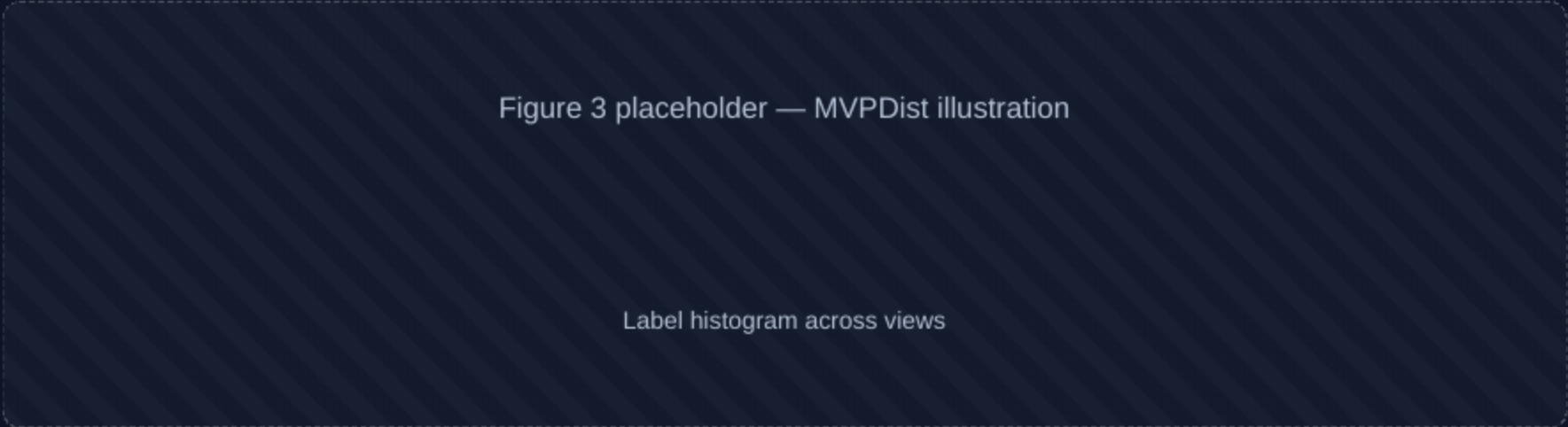
Accelerated Visibility Computation (VAcc)

- Batch GPU projection of all points across frames.
- Compute in-frame and depth-consistency visibilities.
- Aggregate per-mask visibility in one shot; select top-k views.



Multi-View Prompt Distribution (MVPDist)

- Use top-k views; sample LG labels at projected (x,y) for mask points.
- Build per-mask label histogram → class probability.
- Assign class with highest occurrence.



Instance Confidence Scoring

- Score $s_m = s_{IoU} \times s_{class}$.
- s_{class} : max MVPDist probability.
- s_{IoU} : mean IoU between projected 3D mask box and best 2D detected box per view.

Datasets and Setup

- Datasets: ScanNet200 (312 val scenes, long-tail 200 classes) [38], Replica (48 classes) [40].
- Metrics: AP@[0.5:0.95], AP50, AP25.
- Frames: every 10th (ScanNet200), all (Replica). GPU: A100 40GB.
- 2D: YOLO-World-XL (OVOD) [7]. 3D: Mask3D [39] + NMS (no DBSCAN [11]).

RESULTS

Consolidated Results: ScanNet200 and Replica

ScanNet200 (val) — headline speedup					Replica — generalization				
METHOD	MAP	MAP50	MAP25	TIME	METHOD	MAP	MAP50	MAP25	TIME
Open3DIS [34] ...	18.6	23.1	27.3	57.7 s	Open3DIS [34] ...	14.9	18.8	23.6	35.1 s
Open3DIS [34] ...	23.7	29.4	32.8	360.1 s	Open3DIS [34] ...	18.5	24.5	28.2	188.0 s
OpenMask3D [...]	15.4	19.9	23.1	553.9 s	OpenMask3D [...]	13.1	18.4	24.2	547.3 s
Open-YOLO 3...	24.7	31.7	36.2	21.8 s	Open-YOLO 3...	23.7	28.6	34.8	16.6 s
• Head/Common/Tail: 27.8 / 24.3 / 21.6. • ~16× faster vs Open3DIS [34] (2D+3D).									

DISCUSSION

Limitations

- Relies on 3D proposal network only; may miss very small objects due to low 3D resolution.
- 2D fusion methods (e.g., [OVIR-3D \[32\]](#), [Open3DIS \[34\]](#)) can provide richer proposals at higher cost.
- Potential remedy: integrate fast 2D instance segmentation (e.g., [FastSAM \[56\]](#)) to enrich proposals.

Oracle Masks: Prompting Performance

- Closed-vocab [Mask3D \[39\]](#): 35.5 mAP with oracle masks.
- CLIP-based: [OpenMask3D \[42\]](#) 29.1 mAP; [Open3DIS \[34\]](#) 30.9 mAP.
- Our MVPDist: 39.6 mAP (Head 43.3 / Common 36.8 / Tail 38.5).

Numbers from paper Table 5; methods: [Mask3D \[39\]](#), [OpenMask3D \[42\]](#), [Open3DIS \[34\]](#).

Replacing SAM/CLIP and Top-k Views

- [YOLO-World \[7\]](#) crops approach [SAM \[23\]](#) quality with major speedups for [CLIP \[55\]](#) aggregation.
- MVPDist + LG maps: 46.2 mAP with VAcc reduces time to ~17.9 s/scene (Replica GT setup).
- Top-k views (~30–40) improve stability; LG maps ~5× faster than SAM-based HG maps.

Top-k More views → better consensus; 1 view is suboptimal.

Efficient OV 3DIS via 2D Detection + 3D Masks

- Open-YOLO 3D eliminates heavy [SAM \[23\]](#)/[CLIP \[55\]](#) stages.
- MVPDist and VAcc enable fast, accurate text-to-3D mask labeling.
- SOTA accuracy with large speed gains on [ScanNet200 \[38\]](#) and [Replica \[40\]](#).
- Opens path to practical, real-time-capable OV 3D workflows.

References — Baselines and Components

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Project
github.com/aminebdj/OpenYOLO3D